

Sight-Reading Practice Application

INTRODUCTION

Our application aims to solve a problem for learners of the piano. Sight-reading is the act of playing an instrument according to sheet music that the player has never seen or practiced beforehand.

For example, a piano learner might want to sight-read the following passage or similar passages:



An example passage to be sight-read by a beginner pianist.

Without a teacher, it can be difficult for beginner pianists to learn how to sight-read. Learning how to sight-read is important for pianists as it allows them to learn musical pieces faster and play music without memorizing. Really, sight-reading is an important component of musical proficiency for any instrument player. However, not all pianists have access to music teachers to help them develop the skill and provide the feedback they need to improve. This problem of feedback is what our application aims to solve.

As a black box, our solution starts with generating random lines of sheet music (realistic to what a pianist might see in practice) and detecting inputs from a digital piano connected to a computer. As a pianist tries to play according to the sheet music, our software gives visual feedback by changing the colors of notes on the page. After the pianist finishes playing the passage, accuracy scores based on rhythm and pitch are displayed for additional feedback.

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COURSE

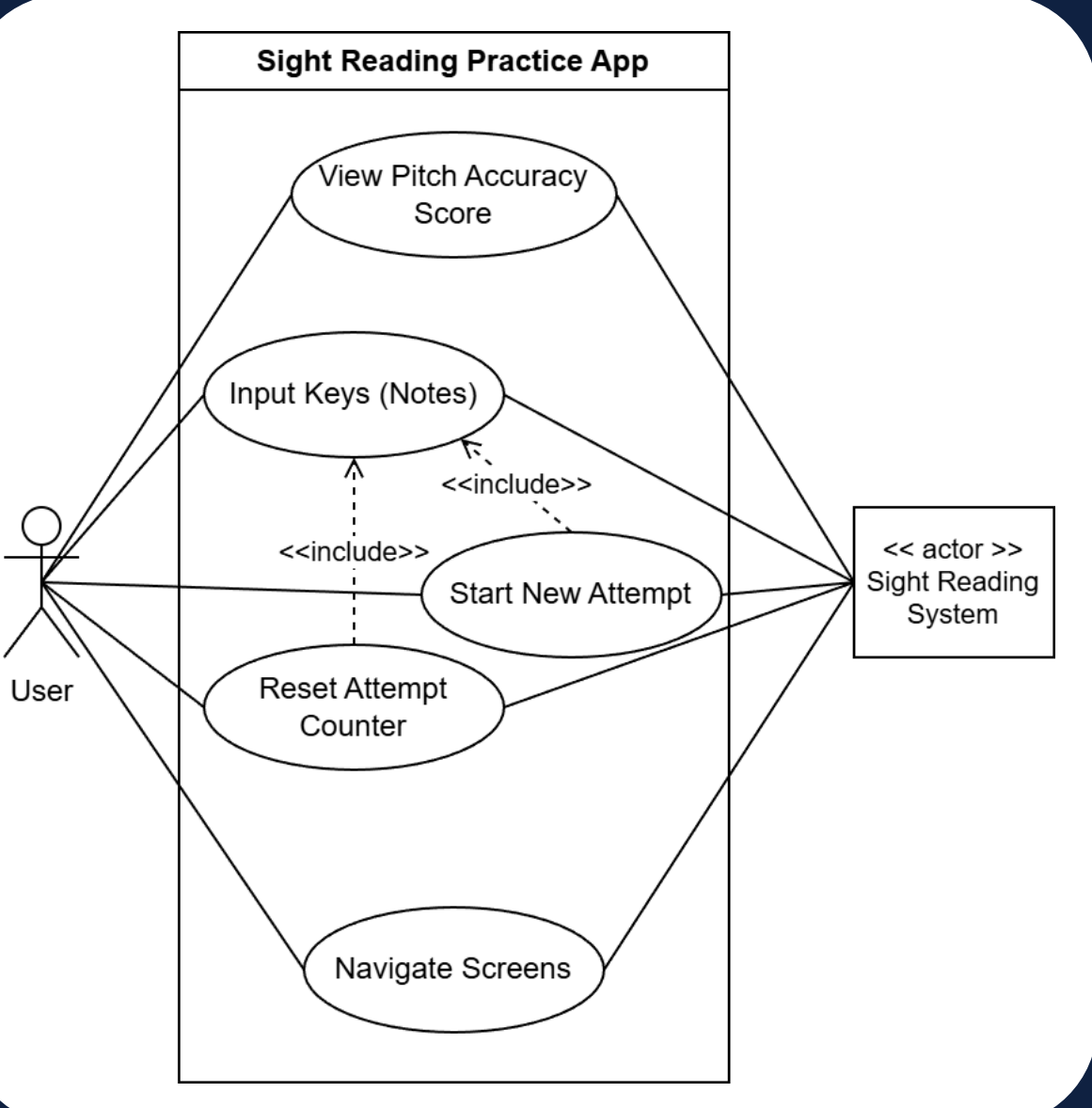
CEN 4010: Software Engineering

Instructor: Dr. Upulee Kanewala

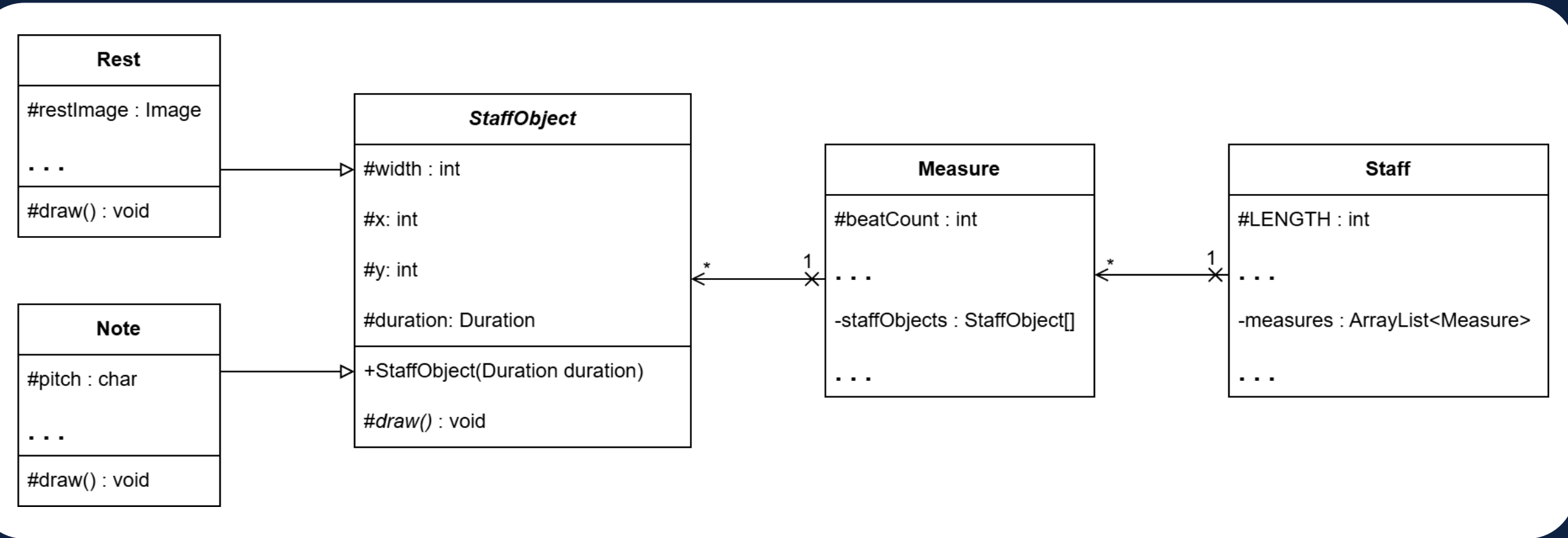


DESIGN

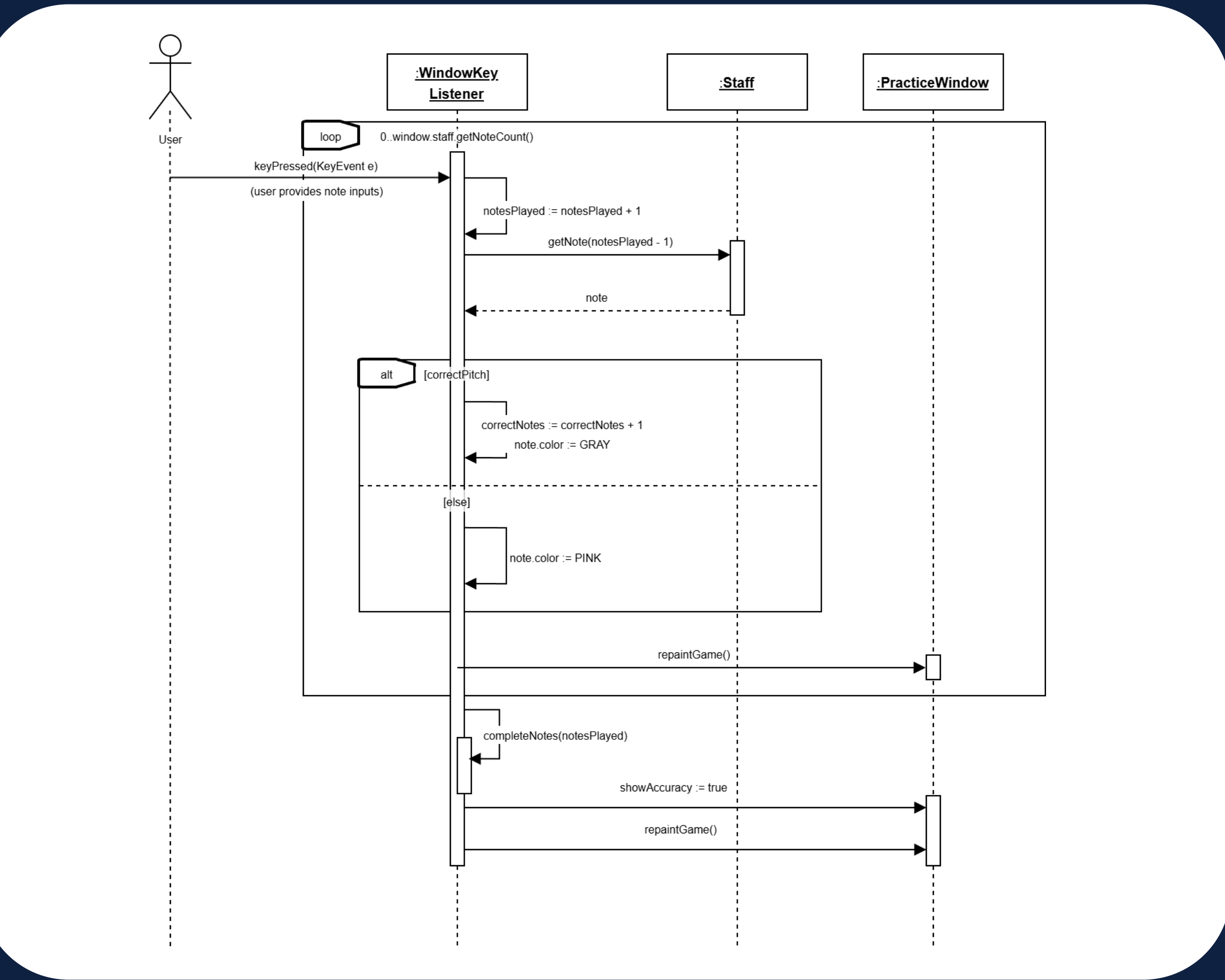
The following diagrams highlight interesting parts of our software’s design (which we may explain further).



Use case diagram for the project.



UML class diagram section showing the design of the staff.



Sequence diagram for the “View Pitch Accuracy Score” use case.

TECHNOLOGIES

Our application was built using the Java programming language using classes from Java’s Abstract Window Toolkit (AWT) and Swing libraries as the basis for the application’s GUI.

The application reads data from a connected digital piano using Java’s MIDI library. MIDI stands for Musical Instrument Digital Interface and is the standard protocol for reading inputs from digital instruments connected to computers. For our program, it importantly allows us to detect when notes are played and the pitches of those notes.

NEXT STEPS

- To finish our course project, we plan to finish implementing the following features:
- Algorithm for random sheet music generation
 - Rhythm accuracy calculation
 - Beaming of shorter note durations within beats
- However, there are many features that should be implemented in the future (that are beyond the scope of our course project) to make the system more useful. The following are a few of these features:
- More sheet music features (dots, ties, key signatures, grand staff, etc.).
 - Customization options (choose key/time signature).
 - Difficulty level option for generated passages.